



FITSTREAK-AI

AI-Powered Fitness Tracking & Personalized Wellness Platform

TECHNICAL PROJECT REPORT

Smart India Hackathon 2026

Problem Statement ID: SIH26196

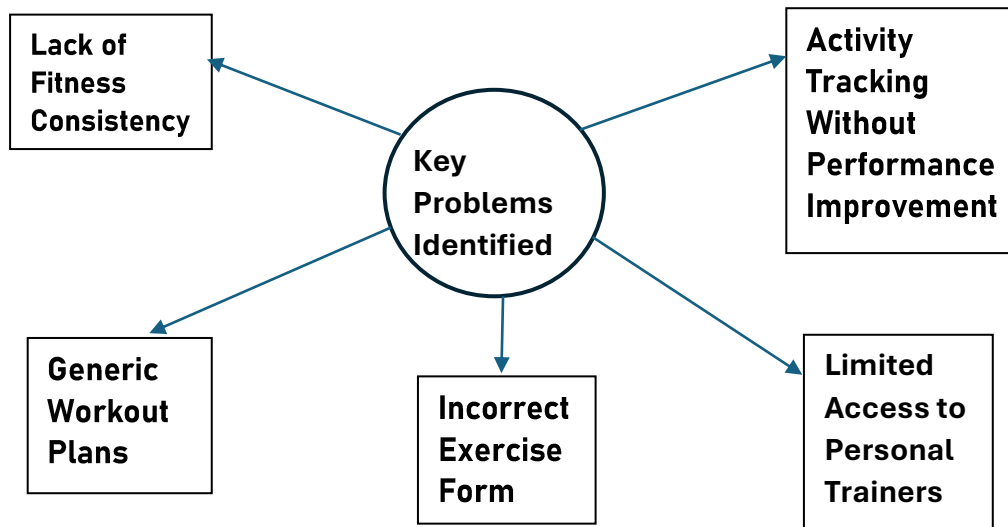
- **Problem Statement:** Student Innovation-Ideas that can boost fitness activities and assist in keeping fit
- **Theme:** Fitness & Sports
- **Category:** Software
- **Team ID:** 175539
- **Team Name:** Innovatrix
- **Team Members :** Ankit Mazumdar (TL)
Ipsita Hazra
Arghya Deep Nag
Ankit Mandal
Abhibrato Chakraborty
Rima Mondal
- **College/Institution :** Asansol Engineering College
- **Academic Year :** 2026-2027

2. Problem Statement & Motivation

2.1 Problem Overview

Maintaining regular physical activity is a major challenge for many students and fitness beginners. Although numerous fitness applications provide workout plans and activity tracking, many users still receive **generic workout recommendations** that do not sufficiently consider their individual fitness level, progress, or exercise performance.

Beginners may also perform exercises with **incorrect posture or form** due to the absence of real-time guidance. At the same time, personal trainers and gyms may not be affordable or accessible to everyone. Traditional fitness trackers mainly focus on recording activities, while providing limited assistance in understanding **how well the activity is being performed**.



2.2 Motivation

The motivation behind **FitStreak-AI** is to bridge the gap between **basic fitness tracking** and **personalized fitness guidance**.

The system combines **computer vision**, **exercise performance analysis**, **progress tracking**, and **adaptive workout recommendations** to create a more personalized fitness experience.

Instead of simply asking:

“Did the user complete the workout?”

FitStreak-AI aims to address:

“How did the user perform, and how can the next workout be improved?”

This approach allows the platform to move from simple activity tracking toward a **performance-oriented and adaptive fitness system**.

★ **FitStreak-AI aims to make fitness more personalized, accessible, interactive, and performance-driven.**

3. Proposed Solution

3.1 Solution Overview

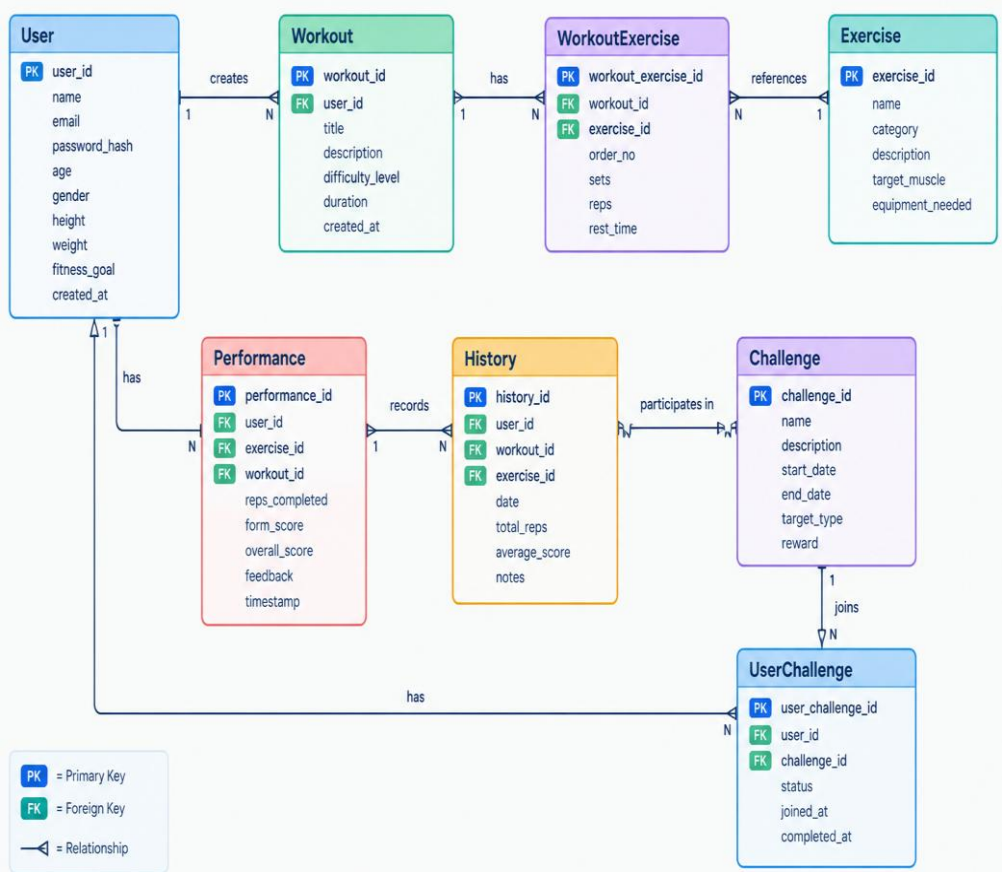
FitStreak-AI is an AI-powered adaptive fitness platform that helps users improve the **quality, consistency, and personalization** of their workouts. It analyzes exercise performance through computer vision, provides real-time feedback, tracks progress, and uses previous performance to adapt future workouts.

What Does FitStreak-AI Solve?

FitStreak-AI addresses the major gaps identified in existing fitness routines:

Existing Problem	FitStreak-AI Solution
Generic workout plans	Personalized workouts based on user performance
No real-time guidance	Exercise analysis & feedback
Incorrect exercise form	Computer-vision based movement analysis
Basic activity tracking	Performance tracking & scoring
Same workout every time	Adaptive future workouts
Lack of motivation	Streaks, challenges & achievements

3.2 ER-Diagram



3.3 Key Objectives

- ❖ **Analyze Exercise Movements:** Evaluates exercise form.
- ❖ **Immediate Performance Guidance:** Gives real-time feedback.
- ❖ **Workouts Based on User Performance:** Tailors routines to user effort.
- ❖ **Adjust Future Workouts:** Scales sessions based on progress.
- ❖ **Monitor Improvement Over Time:** Tracks long-term progress metrics.
- ❖ **Encourage Consistency Through Fitness Activities:** Drives daily habit building via streaks

4. Technology Stack

FitStreak-AI follows a **frontend-backend-database** architecture, with a separate **computer-vision layer** for exercise analysis.

4.1 Frontend — React + Vite

The frontend provides the user interface through which users interact with FitStreak-AI.

Responsible for:

- User dashboard
- Workout selection
- Exercise interface
- Progress display
- Performance results

4.2 Backend — Python + FastAPI

The backend acts as the **core application layer** and manages communication between the frontend, database, and exercise-analysis components.

Responsible for:

- User authentication
- REST API endpoints
- Workout management
- Performance processing
- Storing and retrieving fitness data
- Adaptive workout logic

4.3 Database — PostgreSQL

PostgreSQL is used to securely store and manage the application's fitness-related data.

Stores:

- User profiles
- Workout information
- Exercise performance
- Performance history

4.4 AI & Computer Vision — MediaPipe + OpenCV

The computer-vision layer is responsible for analyzing exercise movements using camera input.

MediaPipe handles:

- Body pose detection
- Body landmark detection
- Joint position information

OpenCV handles:

- Camera/video frame processing
- Image processing
- Supporting real-time exercise analysis

4.5 Adaptive Workout Logic

The backend uses the user's **previous performance and fitness data** to influence future workout recommendations.

Performance Data → Analysis → Stored History → Adaptive Recommendation



5.System Workflow

FitStreak-AI follows a continuous workflow in which user information and exercise performance are processed to provide personalized fitness recommendations.

5.1 Workflow

1. User Registration

The user creates an account and provides basic information.

2. Fitness Profile

Relevant fitness details are used to understand the user's requirements.

3. Workout Selection

The system provides a workout based on the user's profile and available exercises.

4. Camera Input

The user performs the selected exercise using a camera or video input.

5. Pose Detection

MediaPipe identifies body landmarks from the input.

6. Movement Analysis

The detected landmarks are processed to analyze exercise movement.

7. Performance Evaluation

Repetitions and available movement/form-related metrics are evaluated.

8. Performance History

The workout result is stored for progress tracking.

9. Adaptive Recommendation

Previous performance is considered when generating future workout recommendations.



6. AI & Computer Vision

6.1 Overview

FitStreak-AI uses **computer vision** to analyze exercise movements through camera input. Body landmarks are detected and processed to identify movement patterns, count repetitions, and generate performance-related feedback.

6.2 Pose Detection — MediaPipe

MediaPipe detects key body landmarks from the user's camera/video input.

Used for:

- Body landmark detection
- Joint position tracking
- Pose estimation

6.3 Frame Processing — OpenCV

OpenCV processes camera/video frames during exercise analysis.

Used for:

- Video frame processing
- Image processing
- Camera input handling

6.2 Movement Analysis

Detected body landmarks are analyzed to understand the user's **exercise movement and movement cycles**.

6.3 Repetition Counting

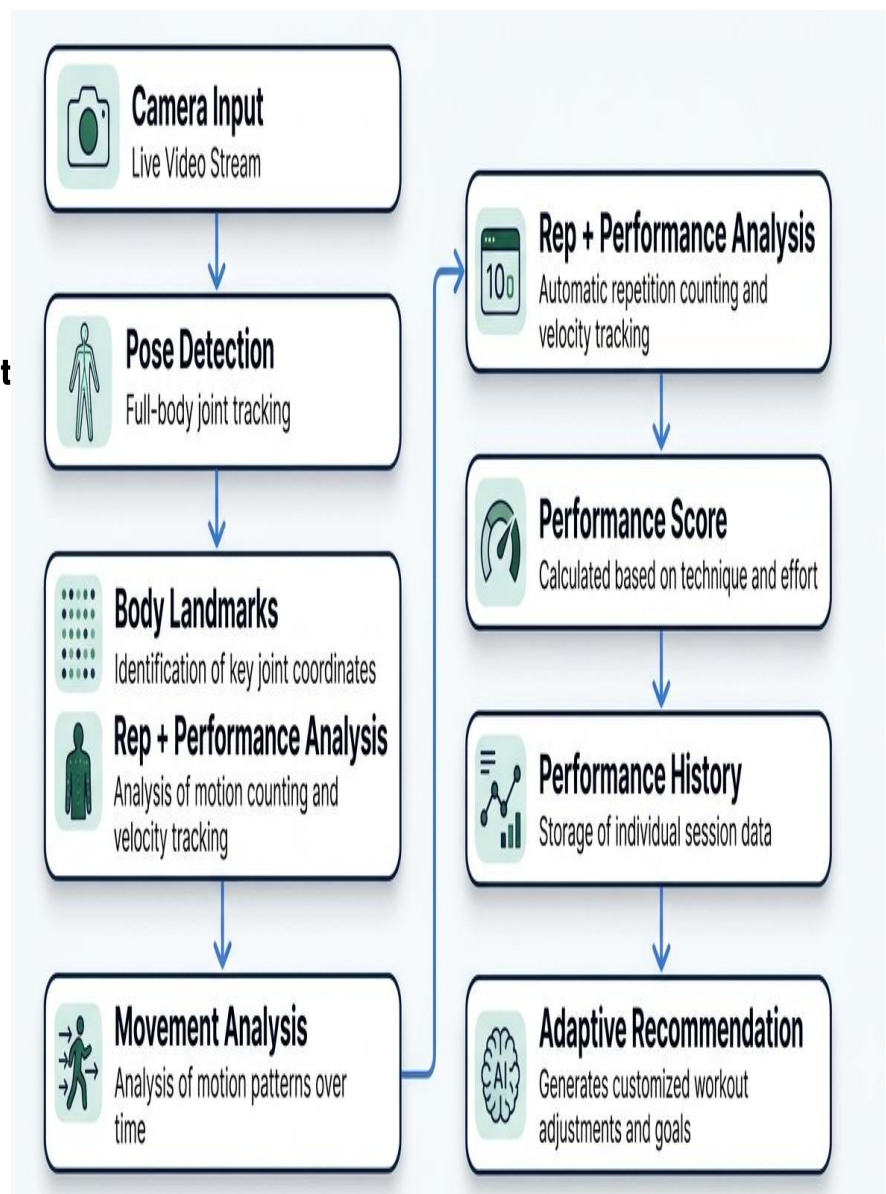
The system analyzes movement patterns to identify **exercise repetitions** during supported exercises.

6.4 Performance Feedback

Exercise metrics are processed to generate **performance-related feedback and scoring**.

6.5 Personalization

The user's **performance history** is stored and used by the application logic to influence **future workout recommendations**.

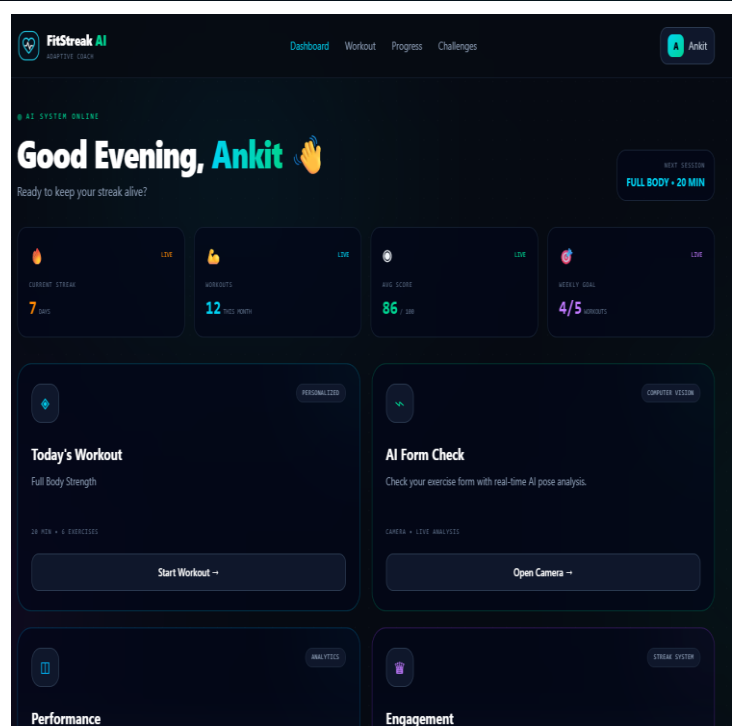
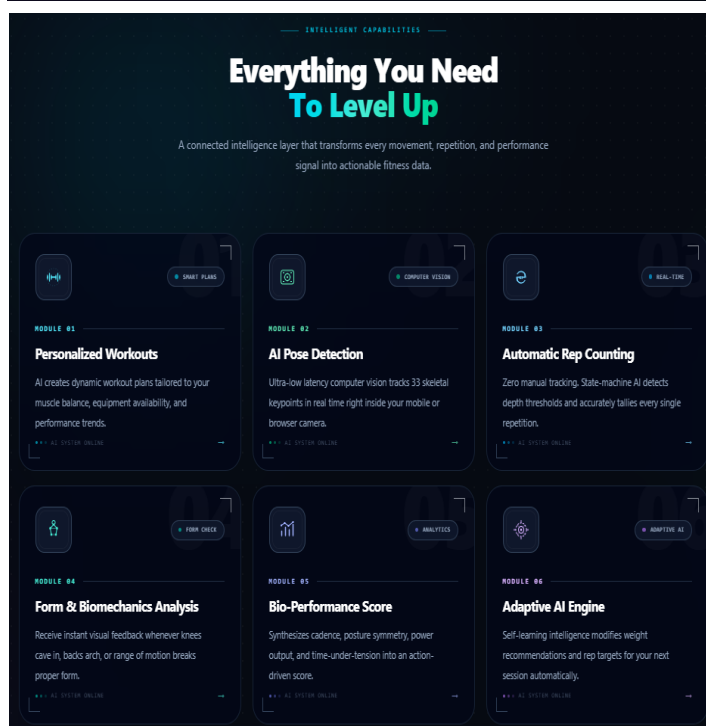
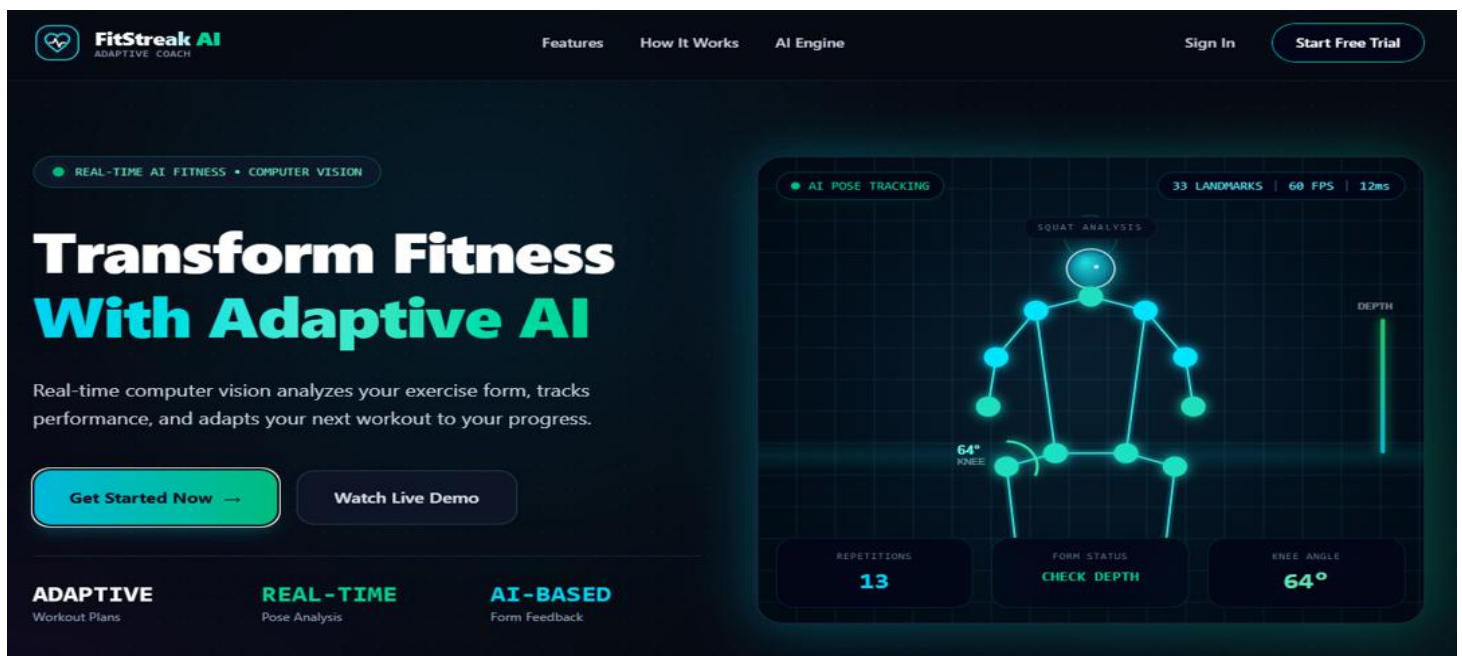


7. Key Features & User Experience

7.1 Key Features

- I. **Home / Landing Page**
Introduces FitStreak-AI and its core fitness features.
- II. **Login / Registration**
Provides secure user authentication and account access.
- III. **Fitness Dashboard**
Displays workout activity, progress and important fitness information.
- IV. **Exercise Analysis**
Uses camera-based movement analysis to monitor exercise performance.
- V. **Performance Result**
Displays exercise results, repetitions and performance-related metrics.
- VI. **Progress / Challenges**
Shows performance history and engagement features such as streaks, challenges or achievements.

7.2 User Experience



8. Feasibility & Impact

8.1 Technical Feasibility

Built using widely available open-source technologies:

- React + Vite — Frontend
- FastAPI — Backend & REST APIs
- PostgreSQL — Data storage
- MediaPipe — Pose detection
- OpenCV — Video processing

Works with a standard device and camera.

8.2 Economic Feasibility

- Uses open-source technologies
- Requires no specialized fitness hardware
- Reduces dependency on continuous personal training

8.3 Operational Feasibility

Register → Workout → Exercise → Feedback → Track → Adapt

8.4 Scalability

- More exercises & workouts
- More users
- Mobile application
- Wearable integration
- Advanced personalization

8.5 Expected Impact

- Consistency — Streaks and challenges encourage regular activity
- Personalization — Workouts adapt to performance
- Guidance — Camera-based exercise feedback
- Performance Awareness — Repetition and performance tracking
- Engagement — Achievements and challenges make fitness interactive

9. Conclusion & Future Scope

9.1 Conclusion

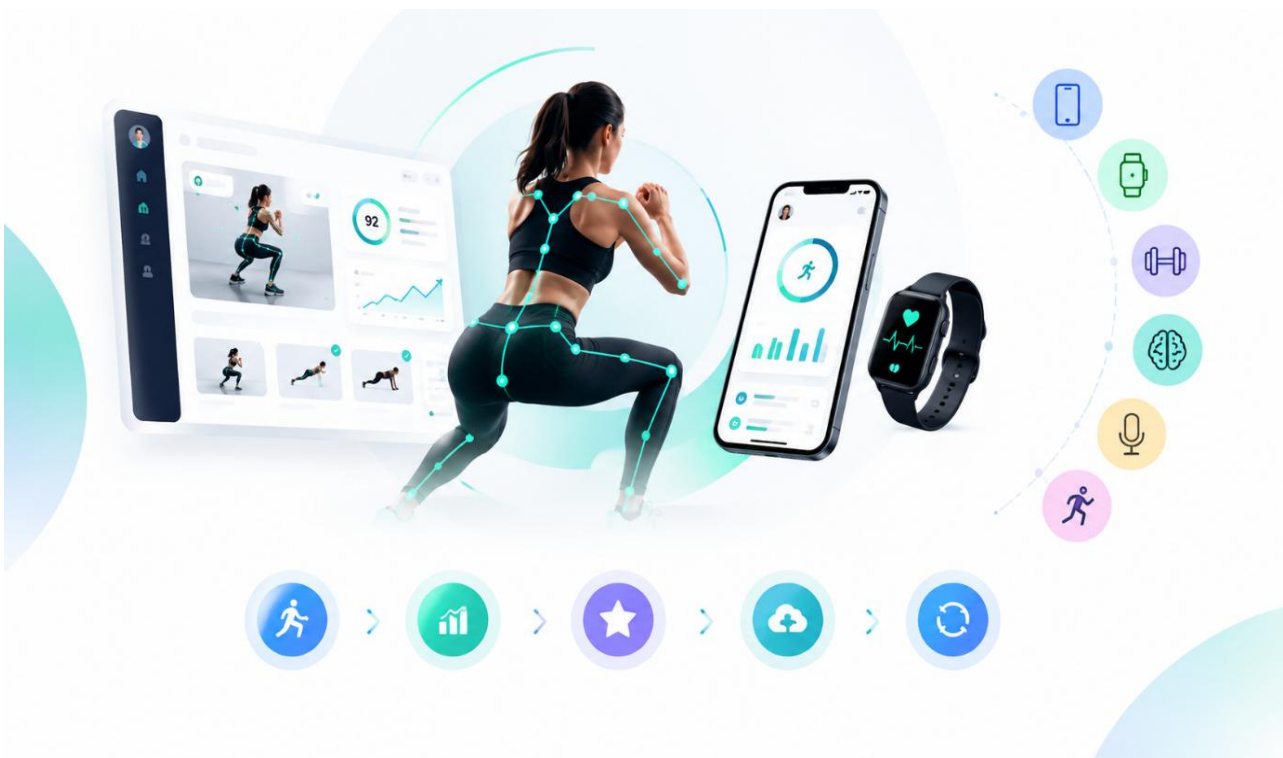
FitStreak-AI combines **computer vision, fitness tracking, performance analysis, and adaptive workout recommendations** to create a personalized fitness experience.

The platform helps users **perform exercises, receive feedback, track progress, and improve future workouts** based on their performance history.

Perform → Analyze → Score → Store → Adapt

9.2 Future Scope

- **Mobile application**
- **Wearable device integration**
- **More exercise categories**
- **Advanced ML-based personalization**
- **Voice-based fitness assistance**
- **Advanced exercise-form analysis**



**SMART INDIA
HACKATHON
2026**

**Team Innovatrix | FitStreak-AI
Smart India Hackathon 2026**